

<h1>  ShieldGig</h1> <h3>Real-Time Income Protection Engine for India's Gig Delivery Workers</h3>

 🏆 Guidewire DEVTrails 2026 — Phase 1 Submission
 👥 Team: [Team Name] 🎯 Persona: Food Delivery Partners (Zomato / Swiggy) </div>

"When it rains heavily, I can't deliver. When there's a curfew, I can't deliver. When the app crashes, I can't deliver. Those days, I earn zero rupees — but my rent doesn't know that." — **Ravi, 26, Zomato delivery partner, Chennai**

⚡ How ShieldGig Works — 15-Second View

1. Rain detected in Ravi's zone → IMD + OpenWeatherMap confirm threshold

2. Claim auto-initiated → zero action from Ravi

3. Fraud check in < 2 seconds → FRS score computed

4. ₹200 credited immediately → ₹100/hr × 2 disruption hours

No forms. No adjusters. No claim ever filed by the worker.

● The Problem

India has **7.7 million** gig delivery workers earning ₹4,000–₹6,000 per week with no paid leave, no sick days, and no safety net. Chennai alone sees ~**80 rain days per year**. On a heavy rain day, a rider loses ₹400–₹600. Cyclone Michaung wiped out 3–4 days of income per worker with no recourse.

Every existing insurance product covers accidents, hospitalization, and death — events that happen rarely. Not one covers the income disruption that happens 80+ days a year.

ShieldGig fixes the right problem.

💡 What ShieldGig Is

ShieldGig is **not an insurance company**. It is an **underwriting intelligence engine** that enables

licensed insurers to profitably serve gig workers — a segment traditional insurance has never been able to reach.

Insurance Partner Model

ShieldGig operates as a technology layer, fully compliant with IRDAI requirements for parametric insurance.

Role	Entity
Risk Underwriter	Licensed insurer — ICICI Lombard / HDFC ERGO
Trigger + Intelligence Engine	ShieldGig
Policy Administration	Guidewire PolicyCenter API
Claims Automation	Guidewire ClaimCenter API
Premium Billing	Guidewire BillingCenter API
Distribution	Zomato / Swiggy platform integration

Guidewire Integration

PolicyCenter

- Weekly policy creation every Monday via PolicyCenter API
- ISS score passed as risk attribute for premium computation
- Policy status synced back to ShieldGig in real time

ClaimCenter

- On trigger: ShieldGig pushes a structured, pre-validated claim payload to ClaimCenter
- Fraud Risk Score attached as a claim attribute — ClaimCenter routes CLEAN claims to auto-approval, FLAGGED to human review queue
- Zero-touch: ClaimCenter → BillingCenter → UPI with no human intervention for clean claims

BillingCenter

- Weekly premium deduction via BillingCenter direct debit scheduling
- Payout disbursement coordinated through BillingCenter's payment gateway

- Worker wallet reconciliation synced weekly

Guidewire Marketplace

- ShieldGig packaged as a Marketplace integration — any insurer on PolicyCenter/ClaimCenter can onboard ShieldGig's parametric trigger engine as a configurable product extension

Parametric Logic — Core Principle

ShieldGig does **not** calculate actual income loss. No investigation needed.

- A measurable disruption index is monitored in real time
- When it crosses a threshold → fixed hourly payout rate fires automatically
- The trigger IS the proof. No claim verification required.

Example:

Rainfall = 72mm/day in Ravi's zone

Threshold = 64.5mm (IMD Heavy Rain classification)

Duration confirmed = 2 hours above threshold

→ Payout = ₹100/hr × 2 hrs = ₹200

→ Settled automatically, no worker action

Persona & Scenarios

Primary Persona: Ravi, Zomato Food Delivery Partner (Chennai)

Attribute	Value
Platform	Zomato (primary), Swiggy (secondary)
Weekly Earnings	₹5,200 (~₹750/day)
Peak Slots	Lunch 12–3:30 PM · Dinner 7–11:30 PM
Device	Android ~₹10,000 budget phone
Annual exposure	~80 rain days · loses ₹400–₹600/heavy rain day

Real Scenario Simulation — Chennai, November Rain

Date: November 12, 2025
Location: Adyar, Chennai
IMD data: 72mm rainfall · threshold crossed for 4 hours

Ravi's plan: Standard Shield (₹49/week)
Trigger tier: Heavy Rain (₹100/hr)

Payout calculation:

₹100/hr × 4 hrs = ₹400 gross

Daily cap check: ₹400 < ₹300 cap → capped at ₹300

Timeline:

11:00 AM → IMD threshold crossed in Adyar zone
11:02 AM → Claim auto-initiated by ShieldGig
11:02 AM → Fraud engine: FRS = 14/100 → CLEAN
11:02 AM → Claim pushed to Guidewire ClaimCenter
11:04 AM → ₹210 (70%) credited to Ravi's UPI
48 hrs → ₹90 (30%) released after review window

Total: ₹300. Premium paid: ₹49. Net benefit: ₹251.

Scenario B — Bandh / Strike (NLP Trigger)

Tuesday 9 AM — local political group announces unplanned shutdown

ShieldGig NLP scraper detects:

News article: "Chennai bandh called in zones 4, 7, 11"

Confidence score: 0.82 → HIGH

Platform API: Zomato deliveries OFFLINE in zones 4, 7, 11

Dual confirmation: PASSED

Workers on Standard Shield in those zones:

→ ₹100/hr payout for verified offline hours
→ Auto-claim pushed to ClaimCenter
→ Zero worker action required

Scenario C — Cyclone Multi-Day

IMD Red Alert + Zomato suspended in coastal zones

Elite Shield workers:

→ ₹150/hr (cyclone rate) × disruption hours/day
→ Daily cap: ₹300/day

- Weekly cap: ₹1,000 total protection ceiling
- Released daily via BillingCenter

End-to-End Workflow

mermaid

graph TD

```
A[Worker opens ShieldGig] --> B[OTP login]
B --> C[Onboarding: platform + zone + avg income]
C --> D[ISS calculated → City risk profile applied]
D --> E[Weekly premium quoted via PolicyCenter]
E --> F{Worker activates?}
F -- Yes --> G[Premium via BillingCenter / UPI Autopay]
F -- No --> H[Push nudge next morning]
G --> I[Policy active 7 days]

I --> J[Background: all triggers polled every 15 min]
J -- No trigger --> K[Week ends → auto-renews]
J -- Trigger detected --> L[Dual-source confirmation check]

L --> M[Hourly payout calculated]
M --> N[Daily cap check ₹300]
N --> O[Weekly cap check ₹1000]
O --> P[Fraud engine: FRS computed]

P --> Q{FRS routing}
Q -- 0-30 CLEAN --> R[Auto-claim → ClaimCenter]
Q -- 31-60 REVIEW --> S[40% provisional + EXIF liveness check]
Q -- 61-100 FLAGGED --> T[Manual review queue in ClaimCenter]

R --> U[70% UPI immediate + 30% after 48hr]
S --> V{Liveness photo verified?}
V -- Yes → instant release --> U
V -- No → fraud confirmed --> W[Clawback + suspension]
```

Weekly Premium Model

Weekly pricing matches the gig worker's income cycle. Premiums deducted Monday morning. Payouts settled Sunday night.

Dynamic Premium Formula

Weekly Premium = Base Rate × ISS Multiplier × City Risk Profile

ISS Multiplier:

ISS 80-100	→	0.85x	(stable, discount)
ISS 50-79	→	1.00x	(standard)
ISS 30-49	→	1.25x	(volatile)
ISS 0-29	→	1.50x	(high risk)

City Risk Profiles — Hyper-Local, Not Generic Tiers

Instead of broad city-tier buckets (Low / Medium / High), ShieldGig builds a **unique risk profile per city** derived from actual local conditions. Two cities can be in the same state and have completely different profiles.

Each city profile is a composite of:

Factor	Data Source	Example Difference
Annual rain days	IMD 10-year station history	Chennai 80 days vs Coimbatore 55 days
Flood zone % of city	NDMA flood hazard maps	Velachery vs Adyar within same city
Political bandh frequency	News archive NLP (past 3 years)	Kolkata high vs Bengaluru low
Platform order density	Mock platform API	Dense cloud kitchen zones vs sparse suburbs
Average disruption hours/event	IMD + historical platform data	Mumbai rain clears in 3 hrs, Chennai in 8 hrs
VVIP movement frequency	State protocol office advisories	Delhi high, Tier-2 cities low

python

Each city gets its own computed multiplier – not a bucket label

```
CITY_RISK_PROFILES = {
    "chennai": {
        "base_multiplier": 1.15,
        "flood_zone_surcharge": {
            "velachery": 0.20,
            "pallikaranai": 0.18,
            "adyar": 0.08,
            "anna_nagar": 0.04,
        },
        "bandh_frequency_score": 0.6,    # moderate
        "avg_disruption_hours": 6.2,
        "platform_density_score": 0.75,
    },
    "mumbai": {
        "base_multiplier": 1.25,
        "flood_zone_surcharge": {
            "dharavi": 0.22,
            "kurla": 0.18,
            "bandra": 0.06,
        },
        "bandh_frequency_score": 0.5,
        "avg_disruption_hours": 4.8,
        "platform_density_score": 0.90,
    },
    "kolkata": {
        "base_multiplier": 1.20,
        "flood_zone_surcharge": {
            "salt_lake": 0.15,
            "howrah": 0.20,
        },
        "bandh_frequency_score": 0.9,    # historically high
        "avg_disruption_hours": 7.1,
        "platform_density_score": 0.65,
    },
    "bengaluru": {
        "base_multiplier": 1.00,
        "flood_zone_surcharge": {
            "koramangala": 0.10,
            "bellandur": 0.14,
            "indiranagar": 0.05,
        },
        "bandh_frequency_score": 0.3,    # low historically
        "avg_disruption_hours": 3.5,
        "platform_density_score": 0.85,
```

```
}  
}  
}
```

This means a worker in Kolkata's Howrah zone pays a meaningfully different premium than a worker in Bengaluru's Indiranagar — not because of a generic tier, but because Howrah floods deeper and Kolkata has more political shutdowns. The pricing is explainable, auditable, and honest.

Plan Tiers

Plan	Weekly Premium	Triggers Covered	Hourly Rate
Basic Shield	₹29	Heavy Rain	₹75/hr
Standard Shield	₹49	Rain + Platform Downtime + Bandh	₹100/hr
Full Shield	₹79	Rain + Platform + Bandh + Heat + Pollution	₹100/hr
Elite Shield	₹109	All triggers + Cyclone	₹150/hr (cyclone)

Scope enforced strictly: Income loss only. Health, accident, vehicle, and life coverage permanently excluded.

⚡ Parametric Triggers & Hourly Rates

Trigger	Source	Fallback	Threshold	Rate	Confirmation
Heavy Rain	IMD station	OpenWeatherMap	≥ 64.5 mm/day	₹100/hr	Dual source
Extreme Rain	IMD	Satellite	≥ 115.6 mm/day	₹150/hr	Dual source
Platform Downtime	Zomato/Swiggy API	Manual flag	> 3 hrs in zone	₹100/hr	Single source
Bandh / Strike	NLP scraper	—	Confidence ≥ 0.6	₹100/hr	NLP + platform offline
VVIP Gridlock	Traffic API	Rider GPS polygon	Speed < 5 km/h 45 min	₹75/hr	Dual source
Severe Heat	IMD	OpenWeatherMap	≥ 43°C 2+ hrs	₹75/hr	Dual source
Severe Pollution	AQICN	State PCB	AQI ≥ 200	₹75/hr	Single source
Cyclone	IMD Red Alert	NDMA advisory	Official alert	₹150/hr	Single source

Liquidity Protection Caps

Explicitly designed to protect the insurance pool during city-wide catastrophes — the scenario where every worker in a zone claims simultaneously.

Per-hour rate	→ varies by trigger type (₹75-₹150/hr)
Daily cap	→ ₹300/day regardless of hours or triggers
Weekly cap	→ ₹1,000/week hard ceiling per worker
Pool reserve	→ 25% of weekly premium pool held in reserve Released only if claims exceed 70% of pool

Why this works for the insurer:

If 10,000 Chennai workers all claim on a cyclone day, the maximum liability is $10,000 \times ₹300 = ₹3,00,00,000$. With a weekly premium pool of ₹49,00,000 and 25% reserve + reinsurance, the insurer's net exposure is calculable and bounded before the event occurs — not discovered after.

AI/ML — What Each Model Does

1. ISS Engine

Scores every worker 0-100 on income stability at onboarding. Determines premium multiplier. Phase 1: Python rule engine. Phase 2: XGBoost trained on zone + income + disruption history data. Adopted only when it outperforms the rule engine by > 5% precision on holdout set.

2. Sunday Night Premium Repricing

Fetches 7-day IMD forecast every Sunday 11 PM. Recalculates next week's premium before the disruption, not after. Cyclone in forecast = premium rises. Clear week = discount applied. Cap: $\pm 40\%$ from base rate.

3. Fraud Detection Engine (FRS 0-100)

Three layers: individual device signal checks (GPS vs cell tower, Wi-Fi fingerprint, battery state, accelerometer) → behavioral baseline deviation → Isolation Forest coordinated ring detection. FRS 0-30 auto-approve. FRS 31-60 provisional payout + EXIF liveness check. FRS 61-100 manual review queue.

4. NLP Disruption Scraper

Polls news feeds every 15 min for bandh/strike/curfew keywords. Scores confidence per article. Never triggers alone — requires platform API to confirm deliveries are offline in the claimed zone. Dual confirmation prevents a distant protest from incorrectly firing Chennai claims.

5. Hourly Payout Calculator

Computes disruption hours from trigger start to end. Applies hourly rate per trigger type. Enforces daily cap (₹300) then weekly cap (₹1,000). Splits output into 70% immediate + 30% held. Result passed to BillingCenter for UPI disbursement.

Adversarial Defense & Anti-Spoofing Strategy

Response to DEVTrails 2026 Market Crash Scenario: 500-worker syndicate used GPS-spoofing to drain the liquidity pool.

Core Principle

GPS is a single data point. We never trust a single data point. A genuine worker stranded in a flood produces a coherent multi-signal profile. A spoofer at home produces an incoherent one.

Four Detection Layers

Layer 1 — Device Sensor Fusion GPS + cell tower triangulation + Wi-Fi BSSID fingerprint + accelerometer motion state + battery charge status. Incoherence across these signals raises FRS.

Layer 2 — Behavioral Baseline Every worker's Personal Activity Graph built over first 2 weeks. Normal zones, typical hours, inferred home location. Claim from an unknown zone = heavy FRS penalty.

Layer 3 — Coordinated Ring Detection Isolation Forest on claim event vectors. If N workers claim the same grid cell in the same 15-minute window with correlated onboarding dates or device subnets → entire cluster flagged. Individual FRS scores overridden by cluster anomaly score.

Layer 4 — Weather Cross-Verification At least one IMD station within 5km must independently confirm the trigger crossed threshold. No corroborating ground station = claim flagged regardless of GPS.

UX Balance — EXIF Liveness Check

Replaces the 72-hour human review wait for borderline claims. When FRS is 31-60 but weather is confirmed:

- 40% provisional payout released immediately
- Worker receives push notification with one-tap camera prompt
- Worker takes a live photo – app reads embedded GPS EXIF data
- EXIF coordinates match claimed zone?
 - YES → remaining 60% released instantly
 - NO → provisional 40% clawed back, account suspended

One photo. Instant resolution. Honest workers are not punished. Spoofers cannot fake EXIF data on a live capture without specialized tools that 99% of syndicate members do not have.

 Income Protection Dashboard

Worker View

This Week

Premium paid: ₹49

Payout received: ₹300 (4-hr heavy rain event, daily cap applied)

Net protected: ₹251 

"Without ShieldGig this week, you would have lost ₹300."

Last 4 weeks:

Wk 1: paid ₹49 received ₹0 clear week

Wk 2: paid ₹49 received ₹300 heavy rain (capped)

Wk 3: paid ₹49 received ₹200 2-hr platform downtime

Wk 4: paid ₹49 received ₹0 clear week

Insurer / Admin View

- Live: active policies, premium pool, loss ratio, fraud flagged count
- 72-hour payout forecast (weather-driven liability with cap projections)
- Fraud review queue → synced to ClaimCenter
- Weekly: premium collected vs payout released (8-week trend)
- Zone risk heatmap colored by per-city disruption profile
- Pool reserve status: current reserve % vs 25% floor

 System Reliability

Failure Mode	Handling
OpenWeatherMap unavailable	Fall back to IMD station feed
IMD data delayed	Last-confirmed reading + 30-min cache
Platform API unreachable	Mock status; admin override
GPS signal lost on device	Last verified location within 15-min window
NLP scraper fails	Trigger held; manual admin review
Single-source trigger only	Held for admin confirmation — no auto-payout

Platform Decision — Mobile PWA (Flutter)

Delivery workers do not use laptops. Every interaction happens on a ₹10,000 Android phone at a red light. Designed for one-thumb operation, 3-second task completion, push notification as the primary channel.

Flutter: single codebase for Android PWA (worker app) + web (insurer admin). Native sensor access for fraud detection. Tamil/Hindi localization.

Tech Stack

Frontend

Component	Technology
Framework	Flutter (Dart) — PWA + Web
State Management	Riverpod
Background Location	flutter_background_geolocation
Local Storage	Hive (offline-first)
Payments (mock)	Razorpay Flutter SDK — test mode

Backend

Component	Technology
API Server	Node.js + Express
Database	Supabase (PostgreSQL + PostGIS for zone queries)
Auth	Supabase Auth (OTP via phone)
Hosting	Render (free tier)
Trigger Polling	Node-cron (every 15 min)
NLP Scraper	Python + spaCy via FastAPI microservice

AI/ML

Component	Technology
ISS Scoring (Phase 1)	Python rule engine via FastAPI
Fraud Anomaly Detection	scikit-learn Isolation Forest
NLP Scraper	spaCy NER + keyword confidence scorer
Premium Repricing	Forecast-weighted rule engine → XGBoost Phase 2

Guidewire

Integration	API
Policy lifecycle	PolicyCenter REST API
Automated claim creation + routing	ClaimCenter REST API
Premium billing + payout	BillingCenter REST API
Distribution packaging	Guidewire Marketplace

External APIs

API	Use	Cost
OpenWeatherMap	Rainfall real-time	Free (1,000 calls/day)
IMD Open Data	Authoritative thresholds + fallback	Free
AQICN	AQI monitoring	Free token
Traffic API	VVIP gridlock trigger	Mock Phase 1
Zomato/Swiggy	Platform status	Mock Phase 1
Razorpay	UPI payout simulation	Test mode

MVP Scope — Phase 1

Proving the parametric loop. Not an enterprise system.

Phase 1 demonstrates:

- Rain trigger via live OpenWeatherMap + IMD

- NLP scraper detecting bandh keywords (mock news feed)
- Hourly payout calculation with daily + weekly caps
- ISS scoring (rule engine)
- Fraud validation: GPS + cell tower + claim velocity
- EXIF liveness check flow (mock)
- Guidewire ClaimCenter payload structure
- UPI payout via Razorpay test mode

Phase 2 adds: full Flutter app · live ClaimCenter/PolicyCenter integration · city risk profile engine
Phase 3 adds: Isolation Forest fraud model · admin dashboard · stress testing · Marketplace packaging

Cost Efficiency

Resource	Cost
All APIs (free tiers)	₹0/month
Supabase	₹0 (500MB free)
Render hosting	₹0 (free tier)
Razorpay test mode	₹0

Total infrastructure: ₹0/month. Zero manual claims cost.

6-Week Plan

Phase 1 (Weeks 1-2) — Current

- ☒ Persona + income research
- ☒ IMD threshold research
- ☒ Hourly payout model + liquidity caps designed
- ☒ NLP scraper architecture
- ☒ City risk profile system designed
- ☒ Fraud + anti-spoofing architecture
- ☒ Guidewire integration mapped
- ☐ Flutter scaffold + Supabase schema
- ☐ ISS rule engine (Python)
- ☐ Guidewire ClaimCenter mock payload

☐ Phase 1 demo video

Phase 2 (Weeks 3-4) — Automation & Protection

- ☐ Full Flutter app — all 5 screens
- ☐ Weather + NLP trigger cron live
- ☐ Hourly payout calculator with caps
- ☐ Auto-claim → ClaimCenter
- ☐ EXIF liveness check flow
- ☐ City risk profiles for Chennai + Mumbai + Bengaluru + Kolkata

Phase 3 (Weeks 5-6) — Scale & Optimise

- ☐ Isolation Forest fraud model
- ☐ Coordinated ring detection
- ☐ Insurer admin dashboard
- ☐ Pool reserve monitor
- ☐ Guidewire Marketplace packaging
- ☐ Final 5-min demo video



Business Viability

Metric	Value
Target workers — Chennai pilot	10,000
Average weekly premium	₹49
Weekly premium pool	₹4,90,000
Pool reserve (25%)	₹1,22,500 held back
Max weekly liability (all claim, hit daily cap)	₹1,00,00,000
Estimated real loss ratio (capped payouts)	~45-55%
Traditional claims cost	₹1,000-₹3,000/claim
ShieldGig claims cost	₹0 (automated)

The daily and weekly caps are what make this actuarially viable. Without them, a single cyclone day could wipe the pool. With them, the maximum liability per worker per week is known in advance.

ShieldGig — Real-time income protection for the workers who keep our cities fed.